

LECTURE-20

Friday.
17/4/26

Population (N)

mean = μ = parameter
SD = σ

Sample (n)

mean = $\bar{x}$ = statistic
SD = s

Confidence Interval:

- As we usually don't know the data of entire population therefore we take a sample from that population. Now we've have into and values like mean etc of that sample which we committed. of population (μ)
- Therefore to find quantities like mean on the basis of sample mean/quantity; we'll use a confidence interval. Here we've an interval which is more practical and tells μ can vary of course.

Formula: Z-test

$$\left(\bar{x} - Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} \right) < \mu < \left(\bar{x} + Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} \right)$$

Point estimate (sample mean) $\bar{x}$ Critical value * Standard Error = Margin error $Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$ SD of Pop. σ Sample size n level of significance (α) = 1 - Confidence level $\rightarrow 95\%$

$$\mu = \bar{x} \pm \text{Margin error}$$

Q: Average Zinc concentration from sample of measurement in 36 diff. locations of river is 2.6 g/ml. Find 95% and 99% confidence interval for the mean concentration in entire ^{population} river. Past study shows that standard deviation is 0.3 g/ml. $\rightarrow$ tells that this is SD of Population (not sample).

$\bar{x} = 2.6$, $n = 36$, $\sigma = 0.3$, Confidence level $\rightarrow 0.95$ & 0.99

• for Confidence level = 0.95

$$\alpha = 1 - 0.95$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$1 - \alpha/2 = 1 - 0.025$$
$$= \boxed{0.975}$$

★ Find this value by reverse track to get corresponding value of Z:

$$Z \text{ at } (1 - \alpha/2) = \boxed{1.96} \rightarrow Z_{\alpha}$$

★ Plug everything in formula.

$$\left[2.6 - (1.96) \left(\frac{0.3}{\sqrt{36}} \right) \right] < \mu < \left[2.6 + (1.96) \left(\frac{0.3}{\sqrt{36}} \right) \right]$$

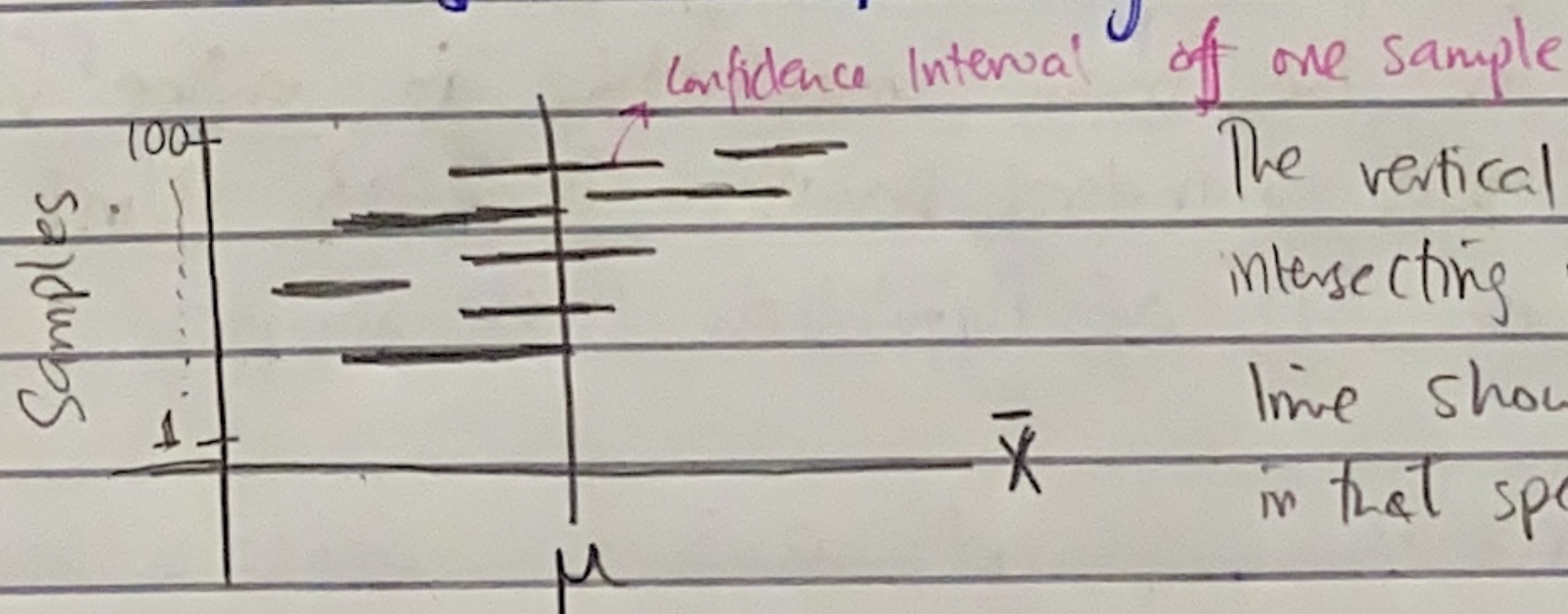
answer:

$$\boxed{2.50 < \mu < 2.70}$$

Interpret:

This does not means that: "there is a 95% chance that actual mean of entire river is b/w 2.50 - 2.70".

This means: "If we repeatedly take 100 samples like this from several locations of river each time; then in 95 samples we'll be having a confidence interval (range of μ) which will actually have the original population mean (μ) in it. 5 samples ki ^{Confidence Interval} range hi bahar hogi asal μ say.



The vertical line of μ intersecting with any horizontal line shows that μ lies in that specific samples C.I.

• for Confidence level = 0.99

$$\alpha = 1 - 0.99$$

$$\alpha = 0.01$$

$$\alpha/2 = 0.005$$

$1 - \alpha/2 = 0.995 \rightarrow$ find corresponding 'Z' value at this.

$$Z_{\alpha} = 2.57$$

• everything else in formula is same so plug values.

$$2.6 - (2.57) \cdot \left(\frac{0.3}{\sqrt{6}} \right) < \mu < 2.6 + (2.57) \left(\frac{0.3}{\sqrt{6}} \right)$$

answer:

$$2.47 < \mu < 2.73$$

Interpret: Since the range of μ increased (lower: 2.5 $\rightarrow$ 2.47) and upper (2.70 $\rightarrow$ 2.73) $\therefore$ this shows that precision has decreased when Confidence coefficient $\uparrow$.

$$\text{Precision} = \text{Upper} - \text{Lower}$$

$$\text{Precision at 95 \%} = 2.70 - 2.50 = 0.20$$

$$\text{precision at 99 \%} = 2.73 - 2.47 = 0.26$$

Q: Confidence interval is 90% for sample size = 225 and it gives μ : 11.15 to 12.05. Now estimate average no. of seats per flight with 99% confidence interval.
 Need to find one confidence interval from another given.

$$C.I = 0.90$$

for

$$11.15 < \mu < 12.05$$

$$n = 225$$

$$\alpha = 1 - 0.90$$

$$= 0.10$$

$$\alpha/2 = 0.05$$

$$1 - \alpha/2 = 0.95$$

$$Z_{\alpha} = 1.645$$

$$\text{lower limit: } \bar{X} - \left(Z_{\alpha} \cdot \frac{s}{\sqrt{n}} \right) = 11.15 \quad (A)$$

$$\text{Upper limit: } \bar{X} + \left(Z_{\alpha} \cdot \frac{s}{\sqrt{n}} \right) = 12.05 \quad (B)$$

$\therefore$ Lower limit + Upper limit = $\boxed{2\bar{X}}$ ($\pm$ Margin error gets cancelled)

$$2\bar{X} = 12.05 + 11.15$$
$$\boxed{\bar{X} = 11.6}$$

★ Now Find ' σ ' by 'Margin Error part' of Eq. (2) (or (1))

$$Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}} = \text{Error} \rightarrow \boxed{\sigma = \frac{\text{Error} \cdot \sqrt{n}}{Z_{\alpha}}}$$

② $\downarrow$

$$\sigma = \frac{12.05 - 11.15}{1.645}$$

$$\boxed{\sigma = 4.1}$$

$\Rightarrow$ Now find Actual C.I with 99% by $\bar{X}$ & σ calculated:

$$\alpha = 1 - 0.99$$
$$= 0.01$$

$$1 - \alpha/2 = 0.995 \rightarrow Z_{\alpha} = \boxed{2.576}$$

$$\mu = 11.6 \pm \frac{(2.576) \cdot 4.1}{\sqrt{225}}$$

$$\boxed{10.9 < \mu < 12.3} \leftarrow \text{answer}$$

□ If we don't have σ (SD of population); we'll see 'n'. If $n \geq 30$: we'll use Z-test and replace σ with S (i.e.: SD of sample given)

If $n < 30$: we'll use T-test

Q: Random sample of 250 components with average life of 840 hours and there variance is 483.

Ans: $n \geq 30$ $\therefore$ use $S = \sqrt{483}$ at σ 's place in formula (Z-test).

□ If population size is known i.e. 'N' and we've a sample size 'n':

Then check if: $n \geq 0.05 * N$, then

sample is large enough to consider a constant with it in the formula $\therefore$ formula for ~~standard error~~ $\uparrow$ becomes.

$\rightarrow$ prev. standard error if $n < 0.05 * N$.

$$\text{Stand. Error} = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$$